

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Winter Examination – 2024

Course: B.Tech

Branch : Petrochemical Engineering

Semester : III

Subject Code & Name: BTPCC303 Stoichiometry

Max Marks: 60

Date: 10 /02/2025

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	1 hp is equal to _____ Watts. a. 725 b. 746 c. 0.75 d. 0.85	Remember	1
2	Unit of pressure in American Engineering system is a. atm b. N/m^2 c. psi d. none of these	Remember	1
3	Equal masses of CH ₄ and H ₂ are mixed in an empty container. The partial pressure of hydrogen in this container expressed as the fraction of total pressure is a. 1/8 b. 0.8888 c. 0.5 d. 0.555	Apply	1
4	Number of gram moles of solute dissolved in 1 liter of solution is called its a. Normality b. Molarity c. Molality d. Volatility	Remember	1
5	Convert 54.75 g/liter of NaCl into molarity a. 2.83 b. 0.94 c. 1.5 d. 0.65	Apply	1
6	The objective of material balance is to assess the a. Input-Output b. Conversion c. losses d. all of these	Understand	1
7	A _____ stream is one that skips one or more stages of the process and goes directly to another downstream stage. a. Recycle b. Bypass c. Purge d. None of these	Understand	1
8	Conversion is always based on _____ a. Limiting Reactant b. Excess Reactant c. On both of these d. None of these	Remember	1
9	Photosynthesis is an _____ reaction. a. Exothermic b. Endothermic c. No heat transfer d. None of these	Understand	1
10	Filling of a tank with fluid is a _____ process. a. Batch b. Semi-batch c. Continuous d. None of these	Understand	1
11	The mass of water vapor per unit mass of bone dry air is called a. Relative Saturation b. Absolute Humidity c. Relative Humidity d. None of these	Remember	1

12	What is the final amount of fluid in the vessel if the system is in steady state? a. Same as the initial amount b. Twice of the initial amount c. Zero d. None of these	Understand	1
Q. 2	Solve the following.		12
A)	i) The flow rate of water through a pipe is reported as 15 cubic feet per minute. Taking density of water as 1 g / cm ³ , calculate the mass flow rate in kg/s . ii) How many grams of carbon is present in 600 g. of CaCO ₃	Apply	6
B)	A suspension of calcium carbonate particles in water flows through a pipe. Your assignment is to determine both the flow rate and the composition by weight of this slurry. You proceed to collect the stream in a graduated cylinder for 1 minute. You then weigh the cylinder, evaporate the collected water and reweigh the cylinder. The following results are obtained. • Mass of empty cylinder = 65 g • Mass of cylinder + collected slurry = 565 g • Volume collected = 455 ml. • Mass of cylinder after evaporation = 215 g Calculate: a) the volumetric flow rate and mass flow rate (per hour) of the suspension. b) The density of the suspension and c) mass fraction of CaCO ₃ .	Apply	6
Q.3	Solve the following.		12
A)	Define Molarity, Normality and Molality. A aqueous solution of sulfuric acid is prepared by dissolving 147 gm of H ₂ SO ₄ in one litre of solution. Find the normality and molarity of the solution.	Understand	6
B)	Cracked gas from petroleum refinery has following composition by volume Methane 45 % , Ethane 25 % , Propane 20% and n-butane 10%. Find:- i. Avg. molecular wt.of mixture ii. Composition by wt. iii . Density of mixture at NTP condition	Apply	6
Q. 4	Solve Any Two of the following.		12
A)	A feed of 100 kmol /h containing 40 mole % A is to be distilled to yield a product containing 95 mole % A and a residue containing 90 mole % B. Calculate the flow rate of distillate and bottom stream. Draw a neat block diagram for the same.	Apply	6
B)	A wet paper pulp contains 75% water. After 100 kg of water removed in a dryer, it is found that the pulp is now containing 30% water. Calculate the weight of the original pulp in kg. Also represent the problem with appropriate block diagram.	Apply	6
C)	Air at a temperature of 25 °C & 760 mm Hg pressure has a relative humidity of 80%. Vapor pressure of water at 25 °C is 20.0 mm Hg. Calculate Absolute humidity, Absolute molal humidity and percentage humidity.	Apply	6

Q.5	Solve Any Two of the following.		12												
A)	With a neat flow diagram , explain the concept of recycle ,bypass and purge in chemical industry. Give appropriate example for each one.	Understand	6												
B)	Vapor phase oxidation of ethyl alcohol using Cu catalyst at 550°C result into formation of acetaldehyde. $\text{C}_2\text{H}_5\text{OH} + 1/2\text{O}_2 \longrightarrow \text{CH}_3\text{CHO} + \text{H}_2\text{O}$ The conversion is 30% and excess air used is 10%. Calculate the composition of product stream if 100 kg of C ₂ H ₅ OH per hour is fed to the reactor.	Apply	6												
C)	i) Define the following terms : Limiting reactant, conversion, yield and selectivity. (You may assume suitable reaction for explanation) ii) Consider the following pair of reactions : $\text{A} \rightarrow 2\text{B} \text{ (desired)}$ $\text{A} \rightarrow \text{C} \text{ (undesired)}$ Suppose 100 mol of a is fed to the batch reactor and the final product contains 10 mol of A , 160 mol of B and 10 mol of C. Calculate i) Fractional conversion of A. ii) Percentage yield of B in terms of A	Analyze	6												
Q. 6	Solve Any Two of the following.		12												
A)	200 kg/h of pure ethylene flowing through a pipe is to heated from 27 °C to 227 °C at a constant pressure. Calculate the heat required to be added for achieving the required outlet temperature. $C_p = 2.83 + 28.601 \times 10^{-3} T,$ where C _p is in kcal/ kmol . K and T in K.	Apply	6												
B)	A heat exchanger for cooling hot hydrocarbon liquid uses 10000 kg /h of cooling water , which enters the exchanger at 21°C.The hot oil at the rate of 5000 kg/h enters at 150°C and leaves at 65°C and has an average specific heat capacity of 2.51 kJ/kg.°C.Calculate the outlet temperature of water. C _p of water = 4.184 kJ/kg.K.	Apply	6												
C)	Tin is melted in an open pan using jacket. The jacket is fed with vapor of eutectic mixture of diphenyl-diphenyl oxide at 171 kPa a. Tin is fed to the pan at 30°C.Calculate the quantity of eutectic mixture condensed per 100kg of tin melted at its melting point temperature. Assume no sub-cooling of vapors. Data for tin: Molecular Weight – 118.7 Melting point --- 231.8°C , Latent heat of fusion – 7201 kJ/kmol Specific heat of solid tin , C _{ms} = 21.14 + 0.02 T C _p is in (KJ/ kmol. K) and T is in Kelvin. Data for Diphenyl- Diphenyloxide mixture at171 kPa a :	Analyze	6												
<table> <tr> <th>Saturation</th><th colspan="3">Enthalpy(kJ/kg)</th></tr> <tr> <th>Temp. °C</th><th>Sensible</th><th>Latent</th><th>Total</th></tr> <tr> <td>282</td><td>522</td><td>278</td><td>800</td></tr> </table>				Saturation	Enthalpy(kJ/kg)			Temp. °C	Sensible	Latent	Total	282	522	278	800
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